



A Dual-Graph Framework for Modelling and Analysis of Movements and Activities

Shahram Payandeh* 

School of Engineering Science, Simon Fraser University, V5A 1S6 Burnaby, Canada

* Correspondence: Shahram Payandeh (payandeh@sfu.ca)

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Abstract: Graph-based representations provide a useful systems-level framework for modelling interactions among structure, dynamics, and behaviour. This paper proposes a dual-graph framework for modelling indoor movements and activities. The first layer is a location graph that represents feasible movement through the spatial connectivity of an indoor environment. The second layer is a mixed causal/contextual activity graph that combines directed activity dependencies with undirected contextual associations. The two layers are coupled through an activity-to-location mapping, yielding a probability-preserving dynamical model in which spatial occupancy is jointly influenced by graph-constrained movement and activity-driven spatial expectations. Two features distinguish the proposed framework from conventional dual-graph models. First, the activity layer is explicitly constructed as a mixed directed/undirected network and second, a cross layer coupled mismatch residual framework is proposed to detect inconsistencies between semantic activity evolution and observed movement. The paper also establishes the probabilistic properties of the movement operator, discusses manual and data-driven construction of the interlayer mapping and introduces an optional reverse-coupling extension. Simulations in a six-location living environment examine the effects of the activity-mixture parameter, the mapping matrix, and the coupling gain. The results support the framework as an interpretable basis for indoor behaviour modelling and also highlight some of its limitations for future studies.

Keywords: Dual-graph modelling; Activity dynamics; Movement modelling; Networked dynamical systems; Indoor behaviour analysis; Residual-based anomaly detection

1 Introduction

Network and graph-based models are central to systems engineering because they provide a common mathematical representation for structure, connectivity, and interactions in complex engineered and natural systems. By abstracting system components as nodes and their relationships as edges, graph models support discrete mathematics, linear algebra, stochastic processes, and spectral analysis. A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of a vertex set $\mathcal{V}$ and an edge set $\mathcal{E}$ that encodes relations among the vertices [1].

Graph-based modelling has found widespread use in robotics, sensing, and networked systems. In multi-robot systems, graphs represent communication topologies, task dependencies, and coordination constraints [2, 3]. In robotic manipulation, graph structures capture relations among objects, contact regions, and end effectors [4, 5]. Sensor networks are similarly represented as graphs whose nodes are sensors and whose edges describe communication or spatial proximity, thereby supporting topology design, routing, and fault analysis [6, 7].

Graphs also support signal and data processing on irregular domains. An irregular domain does not possess a uniform Euclidean grid; nodes can have unequal numbers of neighbours and connectivity is determined by relationships rather than uniform spacing. Graph cuts and normalized cuts are established examples in image segmentation [8, 9]. Graph signal processing (GSP) generalizes filtering, spectral decomposition, and de-noising to signals supported on graph vertices [10, 11].

Graph models have increasingly been applied to human movement and activity. At the biomechanical level, the body can be represented as a skeletal graph for spatio-temporal action recognition [12, 13]. At an environmental level, locations can be represented as nodes and movements as edges, permitting trajectory clustering, routine

discovery, and anomaly detection [14–16]. In multi-person settings, graphs can represent proximity and interaction patterns [17].

1.1 Engineering Motivation and Application Scenarios

The principal application motivating this study is unobtrusive monitoring in the home of an older adult. A practical monitoring system may estimate a person’s room-level location from ambient sensors, wireless sensing, wearable devices, or cameras and depth sensors, while activity-recognition modules estimate semantic states such as preparing a meal, eating, resting, bathing, or sleeping. Engineering value arises not only from estimating these states separately, but from checking whether they are mutually consistent. For example, an inferred bathing activity should be supported by bathroom occupancy, whereas a transition from the bedroom directly to the kitchen should be treated differently from a feasible path through connected rooms. The proposed model therefore targets activity-aware home monitoring, indoor abnormal-behaviour warning, localization-fault detection, and assistive-system decision support.

1.2 Related Work

1.2.1 Indoor activity, localization, and trajectory modelling

Indoor sensing research often treats activity recognition and localization as separate tasks. Some studies have jointly inferred activity and location from wireless fingerprints or have coupled activity with orientation and other contextual variables [18, 19]. Recent graph-based localization work has used sensor fusion, dynamic edge construction, graph neural networks (GNNs), and meta-learning to improve adaptation across indoor environments [20]. These systems demonstrate the value of modelling semantic and spatial information jointly, but their principal objective is classification or localization accuracy rather than an interpretable networked dynamical model linking activity causality, feasible movement, graph filtering, and residual-based diagnosis.

1.2.2 Dual-graph and multilayer models

Dual-graph architectures have been reported in recommendation, spatio-temporal forecasting, electroencephalography (EEG) analysis, robotics, biomechanics, and computer vision. Session recommendation models combine intra-session and inter-session relations [21]; dynamic forecasting systems combine physical and learned connectivity [22]; and EEG models combine anatomical and functional graphs [23]. In robotics and movement analysis, different graphs may represent communication and task relations, kinematic and dynamic relations, or semantic and geometric structure [3, 24, 25]. Multilayer-network theory provides a general basis for representing several edge types and weighted interlayer couplings [26, 27].

1.2.3 Graph filtering and anomaly detection

GSP defines graph Fourier bases and graph filters from a graph operator, commonly an adjacency matrix or graph Laplacian [10, 11]. In indoor movement analysis, Markov models and graph-ranking methods have been used to characterize routine movement and detect deviations [15, 16]. However, anomaly scores are frequently based on a single movement graph or on prediction error without explicitly tracing disagreement between a semantic activity layer and a filtered spatial layer.

Table 1 summarizes representative dual-graph modelling directions using common analytical dimensions.

Table 1. Comparison of representative dual-graph modelling directions using common analytical dimensions

Application Area	Dual-Graph Structure	Interlayer Coupling	Dynamic Modelling	Anomaly Detection
Recommendation and machine learning	Behavioural and intention/semantic graphs	Parallel message passing or feature fusion	Learned latent-state updates	Usually not an explicit objective
Spatio-temporal forecasting	Physical graph and adaptive functional graph	Gating, attention, or learned fusion	Recurrent or temporal GNN dynamics	Prediction error may be used indirectly
Neuroscience	Anatomical and functional/similarity graphs	Feature fusion between graph branches	Windowed or recurrent signal evolution	Usually classification-focused
Robotics and computer vision	Kinematic/geometric, semantic/spatial, or communication/task graphs	Cross-graph message passing	Task-dependent learned dynamics	Often task failure rather than residual diagnosis
Indoor activity and localization	Activity labels and spatial observations	Multi-task or sensor-feature fusion	Classification, tracking, or sequence learning	Limited explicit cross-layer traceability
Proposed framework	Mixed directed/undirected activity graph and spatial location graph	Explicit matrices B and optional C	Probability-preserving networked state model	GSP-filtered cross-layer mismatch residual with traceability

Note: GSP = Graph signal processing; GNN = Graph neural network.

1.3 Research Gaps and Contributions

The literature reveals some specific gaps. For example, a single-graph movement model cannot distinguish spatial feasibility from the semantic dependencies that motivate movement. Many dual-graph architectures use two graph branches that do not explicitly represent directed causal activity relations together with undirected contextual relations in one mixed activity layer. Through such mixed layers it is possible to construct a mechanism for detection and localization of inconsistencies between activity and movement. This paper contributes to the construction of a mixed causal-contextual activity operator combined with a spatial movement graph. It offers an explicit activity-to-location mapping with manual-prior and data-driven weight-assignment alternatives. It demonstrates how a probability-preserving coupled dynamical model and GSP framework can offer residual-based anomaly detection with quantitative performance metrics.

2 Problem Formulation and Dual-Graph Model

In the following calligraphic symbols denote sets and graphs, bold upper-case symbols denote matrices, and bold lower-case symbols denote column vectors. Let $\mathcal{G}_L = (\mathcal{V}_L, \mathcal{E}_L)$ be the location graph with $N_L = |\mathcal{V}_L|$, adjacency matrix A_L , degree matrix D_L , and the Laplacian as:

$$L_L = D_L - A_L. \quad (1)$$

The location state $x_t \in \mathbb{R}^{N_L}$ is a column probability vector satisfying $x_t \geq 0$ and $1^\top x_t = 1$.

Let the mixed activity graph be defined as:

$$\mathcal{G}_A = (\mathcal{V}_A, \mathcal{E}_A^{(d)}, \mathcal{E}_A^{(u)}).$$

Its directed and undirected weighted adjacency matrices are A_A and U_A , respectively. Under the adopted source-to-destination convention, entry (i, j) represents influence or transition from node i to node j .

2.1 Mixed Causal-Contextual Activity Operator

The adjacency activity matrices defined above are row-normalized so that every row has unit sum. We denote the resulting matrices by $\bar{A}_A$ and $\bar{U}_A$. The composite activity interaction matrix is:

$$M_A(\beta) = \beta \bar{A}_A + (1 - \beta) \bar{U}_A, \quad 0 \leq \beta \leq 1. \quad (2)$$

Because M_A is a combination of non-negative row-stochastic matrices, it is also non-negative and row-stochastic:

$$\sum_{j=1}^{N_A} (M_A)_{ij} = \beta \sum_{j=1}^{N_A} (\bar{A}_A)_{ij} + (1 - \beta) \sum_{j=1}^{N_A} (\bar{U}_A)_{ij} = \beta + (1 - \beta) = 1. \quad (3)$$

This weighted fusion follows the basis for multilayer-network principle that relation-specific operators can be combined through interpretable layer weights [26, 27].

The activity state $a_t \in \mathbb{R}^{N_A}$ evolves according to

$$a_{t+1} = M_A^\top a_t + \eta_t, \quad (4)$$

Eq. (4) is a first-order networked dynamical approximation: the current activity distribution propagates through graph-defined influences, while η_t can be interpreted as spontaneous choices, unmodelled context, and estimation uncertainty [28, 29].

2.2 Movement Dynamics and Construction of the Transition Matrix

Let $\mathcal{N}(i)$ be the set of locations adjacent to location i , let $d_i = |\mathcal{N}(i)|$ its degree, and let $s_i \in [0, 1]$ be its dwell probability. Define

$$P_{ij} = \begin{cases} s_i, & j = i, \\ \frac{1 - s_i}{d_i}, & j \in \mathcal{N}(i), \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

For any row i ,

$$\sum_{j=1}^{N_L} P_{ij} = P_{ii} + \sum_{j \in \mathcal{N}(i)} P_{ij} = s_i + d_i \frac{1 - s_i}{d_i} = 1. \quad (6)$$

The probability-preserving movement update is:

$$x_{t+1} = P^\top x_t + \epsilon_t, \quad (7)$$

2.3 Activity-to-Location Mapping

The matrix $B \in \mathbb{R}^{N_L \times N_A}$ maps activity states into spatial expectations. Its entries satisfy $B_{ij} \geq 0$, and each activity column is normalized so that $\sum_i B_{ij} = 1$. The activity-induced spatial distribution is:

$$\hat{x}_t^{(A)} = \frac{Ba_t}{\mathbf{1}^\top Ba_t}, \quad (8)$$

provided the denominator is positive.

This framework can support two weight-assignment schemes. For example, in a manual/prior scheme, B_{ij} is assigned from room-function knowledge: a value of one can identify the primary room of an activity, while smaller values represent plausible secondary rooms. This approach is appropriate when data are scarce and domain knowledge is reliable (e.g. care-giver's knowledge). In a data-driven scheme, weights are estimated from synchronized activity and location observations.

2.4 Cross-Layer Coupling

The coupled location dynamics are defined as:

$$x_{t+1} = (1 - \lambda)P^\top x_t + \lambda \hat{x}_t^{(A)} + \epsilon_t, \quad 0 \leq \lambda \leq 1. \quad (9)$$

In the absence of noise, both terms are probability vectors, so their combination remains a probability vector.

The gain λ has an interpretable meaning: $\lambda = 0$ yields topology-driven movement, whereas $\lambda = 1$ makes the next spatial state entirely activity-driven.

The baseline model assumes that activity influences location but location does not directly change activity. With the augmented state $z_t = [a_t^\top, x_t^\top]^\top$, a linearised, unnormalized representation is:

$$z_{t+1} = \begin{bmatrix} M_A^\top & 0 \\ \lambda B & (1 - \lambda)P^\top \end{bmatrix} z_t + \begin{bmatrix} \eta_t \\ \epsilon_t \end{bmatrix}. \quad (10)$$

The zero upper-right block is a modelling baseline, not a universal behavioural claim. It is reasonable when the activity estimator already incorporates environmental context or when the study seeks to isolate semantic-to-spatial influence.

For settings in which location constrains activity selection, define $C \in \mathbb{R}^{N_A \times N_L}$ and feedback gain $\gamma \in [0, 1]$:

$$a_{t+1} = (1 - \gamma)M_A^\top a_t + \gamma Cx_t + \eta_t, \quad (11)$$

$$x_{t+1} = (1 - \lambda)P^\top x_t + \lambda Ba_t + \epsilon_t. \quad (12)$$

The corresponding block matrix contains both off-diagonal blocks, γC and λB . However, the baseline is easier to identify and analyse; the feedback model is more expressive but requires additional data and further analysis.

2.5 Graph Filtering

The observed spatial signal is filtered before cross-layer comparison:

$$\tilde{x}_t = H(L)x_t. \quad (13)$$

For example, for the Laplacian L_L defined previously, the heat-kernel filter can be utilized:

$$H_c = \exp(-\tau L_L), \quad \tau > 0. \quad (14)$$

Additionally, one can also define a normalized alternative based on:

$$L_{\text{sym}} = I - D_L^{-1/2} A_L D_L^{-1/2}, \quad H_n = \exp(-\tau L_{\text{sym}}). \quad (15)$$

An interpretation of the normalized filter is to compensate for degree heterogeneity and prevent high-degree rooms, such as a central living room, from dominating spectral smoothing.

The vector residual and scalar mismatch residual can further be defined as:

$$e_t = \tilde{x}_t - \hat{x}_t^{(A)}, \quad r_t = \|e_t\|_2. \quad (16)$$

A large r_t indicates cross-layer (causal/temporal) inconsistency. The vector e_t provides traceability. For example, the normalized contribution of location i is:

$$q_{i,t} = \frac{e_{i,t}^2}{\sum_{k=1}^{N_L} e_{k,t}^2}, \quad (17)$$

when $r_t > 0$. Locations with the largest $q_{i,t}$ identify where the disagreement is concentrated. A positive $e_{i,t}$ means observed occupancy exceeds the activity-based expectation; a negative value means the activity model expects more occupancy than is observed. This distinction helps separate a spatial jump or localization error from an activity-label conflict. Temporal persistence, topology violations, and the dominant positive/negative residual components are jointly inspected for case-level interpretation.

3 Illustrative Three-Room Example

Consider three locations, Kitchen (K), Living Room (L), and Bedroom (B), connected as $K-L-B$. With node order (K, L, B) ,

$$A_L = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad L_L = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

For dwell probability $s = 0.6$,

$$P = \begin{bmatrix} 0.6 & 0.4 & 0 \\ 0.2 & 0.6 & 0.2 \\ 0 & 0.4 & 0.6 \end{bmatrix}.$$

Each row sums to one. For column-state propagation, an initial Kitchen state $x_0 = [1, 0, 0]^\top$ gives

$$x_1 = P^\top x_0 = [0.6, 0.4, 0]^\top,$$

which preserves probability mass and has the expected 0.4 probability of moving from the Kitchen to the Living Room.

Let the six activities be Prepare (Pr), Clean (Cl), Read (Rd), Watch TV (TV), Sleep (Sl), and Dress (Dr), with two activities assigned to each room. For $\beta = 0.7$, the fused outgoing weights from Pr are 1.0 toward Cl and 0.7 toward Rd. Row normalization therefore produces transition probabilities 0.588 and 0.412. If $a_0 = [1, 0, 0, 0, 0, 0]^\top$, then

$$a_1 = M_A^\top a_0 = [0, 0.588, 0.412, 0, 0, 0]^\top.$$

With

$$B = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix},$$

the activity-induced spatial prediction is

$$\hat{x}_1^{(A)} = [0.588, 0.412, 0]^\top.$$

The nominal raw mismatch is consequently

$$\|x_1 - \hat{x}_1^{(A)}\|_2 \approx 0.017,$$

indicating close agreement. If a sensor instead reports Bedroom occupancy, $x_1^{(\text{obs})} = [0, 0, 1]^\top$, the mismatch becomes approximately 1.231. The component residual attributes the disagreement primarily to unexpected Bedroom occupancy and missing Kitchen/Living-Room support, illustrating both detection and traceability.

3.1 Illustrative Simulation Study

The simulated home contains six locations: Kitchen, Dining Room, Living Room, Bedroom, Bathroom, and Patio. Edges are Kitchen-Dining Room, Kitchen-Bathroom, Dining Room-Living Room, Living Room-Bedroom, Living Room-Patio, and Bedroom-Bathroom. Fourteen activities are assigned: prepare meal, clean dishes, and make tea in the Kitchen; eat and read mail in the Dining Room; watch television, read a book, and rest in the Living Room; sleep and dress in the Bedroom; brush teeth and bathe in the Bathroom; and sit outside and water plants on the Patio.

All time-dependent plots report the discrete simulation step on the horizontal axis. The activity-state and location-state heatmaps use probability values on a fixed colour scale from (0) to (1). Unless a parameter is explicitly varied, the representative nominal simulation uses ($T = 150$) time steps, stay probability ($s = 0.65$), activity-mixture weight ($\beta = 0.60$), activity-to-location coupling gain ($\lambda = 0.50$), reverse-feedback gain ($\gamma = 0$), heat-kernel parameter ($\tau = 0.60$), activity-noise standard deviation (0.010), spatial process-noise standard deviation (0.010), and sensor-noise standard deviation (0.015). The parameter-sweep and quantitative anomaly-detection experiments use ($T = 120$) time steps per trial.

3.2 Effect of β

The parameter $\beta \in [0, 1]$ controls the relative contribution of directed causal transitions and undirected contextual associations in the mixed activity operator. Specifically, $\beta = 0$ produces an operator governed by contextual activity relationships, whereas $\beta = 1$ produces an operator governed by directed causal or sequential relationships. After adding the self-persistence term, the resulting operator is row-normalized to obtain a stochastic activity-transition matrix.

Figure 1a reports the effect of β on the semantic mismatch residual between the graph-filtered spatial observation and the activity-predicted spatial state. For each value of β , 12 independent simulations of 120 discrete time steps were performed. Figure 1b complements this parameter study by illustrating the temporal evolution of the activity-state probabilities during one representative nominal simulation at $\beta = 0.60$. Thus, the heatmap should be interpreted as an illustrative baseline example rather than as a direct visualization of activity redistribution over all values of β .

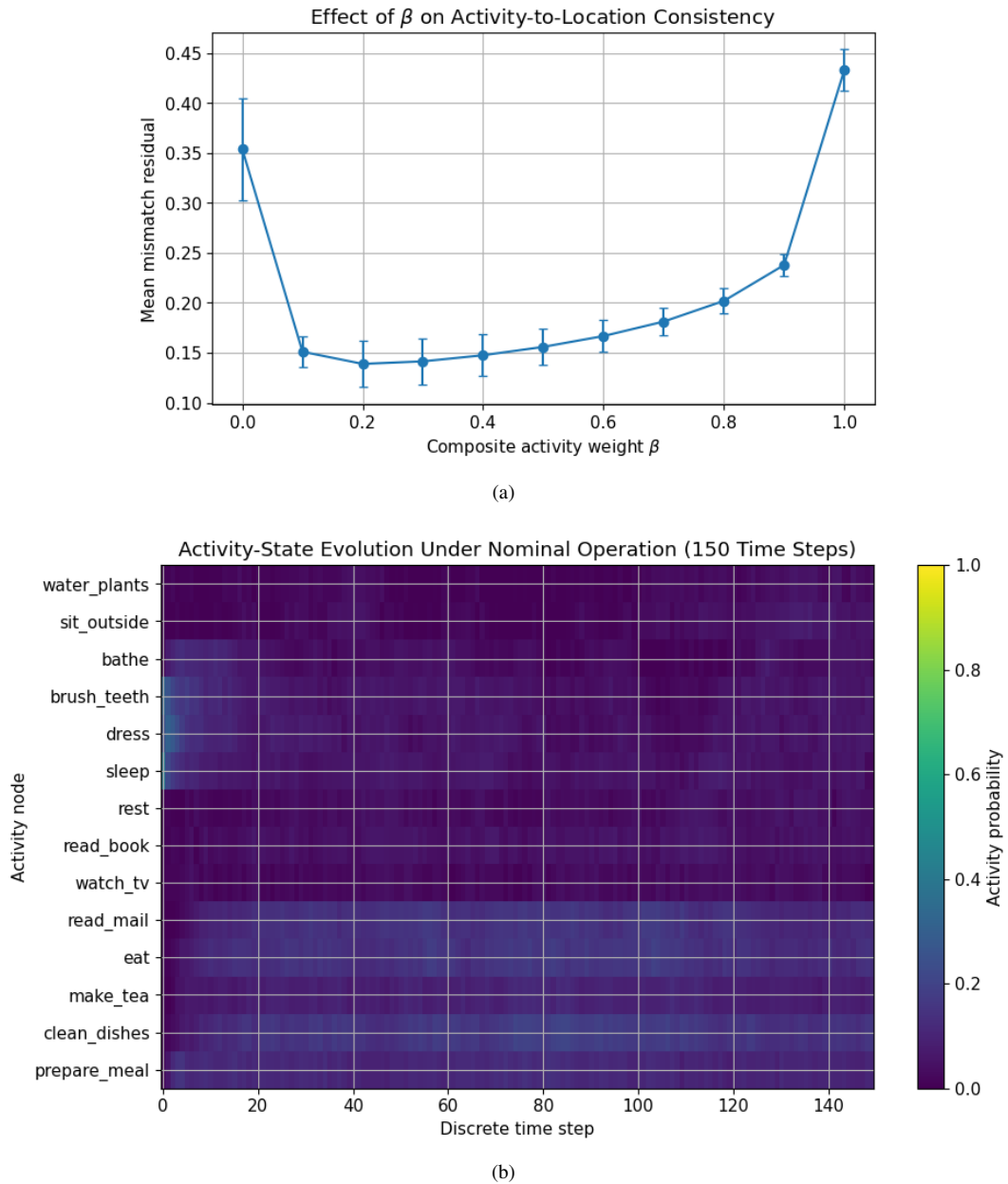


Figure 1. Variation in residual magnitude as a function of β and example of activity state evolution: (a) Mean mismatch residual versus β . Horizontal axis: β ; vertical axis: mean r_t over 120 steps; (b) Activity-state probability over 120 simulation steps. The colour bar spans probability 0 to 1

3.3 Effect of λ

The coupling gain $\lambda \in [0, 1]$ controls the relative contributions of topology-driven movement and activity-driven spatial prediction to the location-state dynamics. In the probability-preserving coupled model, the term $P^T x_t$ in the coupled-location equation represents the spatial state predicted from the location-transition topology, whereas $\hat{x}_t^{(A)}$ represents the spatial state inferred from the current activity distribution through the activity-to-location mapping. Figure 2a evaluates the effect of λ using two complementary residual measures. The semantic mismatch residual, $r_t^{(A)} = \|\tilde{x}_t - \hat{x}_t^{(A)}\|_2$, measures disagreement between the graph-filtered observed location state and the activity-predicted spatial state. The movement-innovation residual, $r_t^{(M)} = \|\tilde{x}_t - P^T x_t\|_2$ which measures disagreement between the filtered observation and the topology-only movement prediction. For each value of λ , 12 independent simulations of 120 discrete time steps were performed using the baseline activity-mixture weight $\beta = 0.60$ and the soft activity-to-location mapping. Figure 2b complements the parameter sweep by showing the temporal evolution of the sensor-observed location-state probabilities for one representative nominal simulation at the baseline value $\lambda = 0.50$.

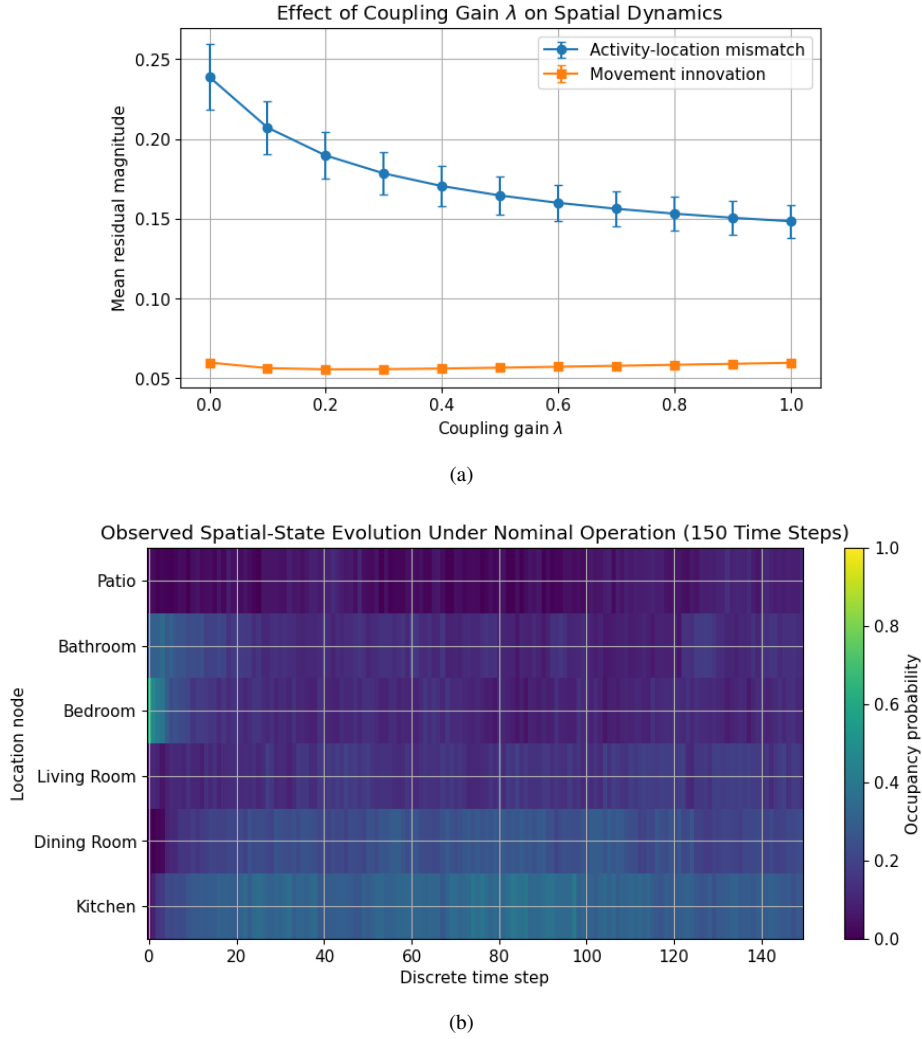


Figure 2. Spatial consistency and occupancy state under varying coupling gain λ

(a) Effect of the activity-to-location coupling gain λ on semantic and movement consistency. The horizontal axis gives λ , and the vertical axis gives the time-averaged residual. The activity-location mismatch curve reports $\|\tilde{x}_t - \hat{x}_t^{(A)}\|_2$, whereas the movement-innovation curve reports $\|\tilde{x}_t - P^T x_t\|_2$. Each point is the mean of 12 independent simulations, each containing 120 discrete time steps. Error bars indicate one standard deviation across the trials; (b) Sensor-observed location-state probability during a representative nominal simulation of 150 discrete time steps at $\lambda = 0.50$. The horizontal axis gives the discrete simulation step, the vertical axis lists the six location nodes, and the colour scale represents location probability from 0 to 1.

3.4 Quantitative Anomaly Evaluation

To evaluate the anomaly detector more objectively, the residual signals were tested using repeated Monte Carlo simulations rather than relying only on visual inspection of individual plots. The detector combines the semantic mismatch residual, which measures disagreement between the observed location and the activity-predicted location, with the movement residual, which measures disagreement with the expected transition over the location graph. A detection threshold of $\theta = 0.3071$ was estimated from 30 nominal simulations using the 99th percentile of the nominal anomaly scores. The independent test set contained 15 nominal sequences and 45 abnormal sequences, including semantic activity-location conflicts, non-adjacent spatial jumps, and localization-sensor corruptions. Each sequence contained 120 discrete time steps, and a time step was classified as abnormal when its combined residual exceeded the threshold. The detector achieved an accuracy of 0.9772, precision of 0.5941, recall of 1.0000, $F1$ score of 0.7453, and receiver operating characteristic (ROC) area under the curve (ROC-AUC) of 0.9996. Recall indicates the fraction of abnormal samples that were detected, while precision indicates the fraction of generated alarms that corresponded to true anomalies. The $F1$ score balances precision and recall. The ROC curve evaluates the detector over all possible threshold values by comparing the true-positive rate with the false-positive rate, and ROC-AUC summarizes this curve as a value between 0.5 for chance-level discrimination and 1 for perfect separation. The results show that all injected anomalies were detected in the controlled simulation, although some nominal samples produced false alarms. These results validate the internal operation of the proposed residual-based framework but should not be interpreted as expected performance in a real occupied home.

4 Model Limitations

The current study has several limitations. First, the main formulation assumes static location and activity graphs; architectural changes, temporary obstacles, and time-dependent routines are represented only through parameter variation. Second, the model describes one monitored person and does not include social interaction, shared-room occupancy, or identity ambiguity. Third, the activity graph and manual form of B depend on expert prior knowledge and can encode incorrect assumptions. Fourth, the linear dynamics are first-order approximations and may not represent long-duration activities, non-Markovian routines, or abrupt context changes. Fifth, the quantitative evaluation uses synthetic data and controlled anomaly injection; real sensing systems will introduce missing observations, class imbalance, sensor drift, and uncertain ground truth. Finally, residual magnitude alone cannot establish clinical significance and must not be interpreted as a diagnosis.

5 Conclusions and Future Work

This paper presented a dual-graph framework that separates indoor behaviour into a spatial movement layer and a mixed causal-contextual activity layer. The study formalized the matrix conventions, established stochasticity and probability preservation, distinguished manual and learned interlayer mappings, and introduced a feedback-coupled extension. The GSP component was expanded to compare combinatorial and normalized Laplacians, and the scalar mismatch residual was augmented by a location-level traceability measure.

Example parameter studies demonstrate how β and λ regulate semantic evolution, spatial prediction, and cross-layer consistency. The Monte Carlo study provides an initial quantitative validation of residual-based anomaly detection under controlled semantic conflicts, topology violations, and sensing errors. The framework is therefore useful not merely as a trajectory simulator, but as an interpretable systems model linking activities, movements, filtering, and anomaly diagnosis.

Future work can follow four implementable routes. First, M_A and B will be learned from synchronized activity and location data. For example, a constrained GNN regression model can predict edge weights where the learned model can be compared with manual priors through likelihood and residual-detection performance. Second, static graphs will be replaced by temporal variational graph models. Time-indexed edges can be generated by a recurrent variational encoder, producing $M_{A,t}$ and P_t with uncertainty intervals. This would represent changing accessibility, daily routines, and context-dependent activity-location relations. Third, anomaly detection can be extended through statistically calibrated thresholds in which residual contributions in the residual-contribution expression can be combined with topology-violation and activity-confidence scores to distinguish behavioural deviations from sensor faults. Fourth, the model will be validated using synchronized smart-home sensors, wearable inertial measurements, wireless channel-state information (CSI), and RGB-D observations. Multi-person extensions will add an interaction graph and a data-association layer. These studies will test generalization across homes and residents and will examine privacy-preserving, edge-computing implementations suitable for older-adult monitoring.

Data Availability

Not applicable.

Conflicts of Interest

The author declares no conflicts of interest.

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